FURTHER MECHANICS / KINEMATICS & DYNAMICS

Impulse and Work Done by a Variable Force

Worked solutions 11 questions · 119 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*6 marks*	2
Question 2	*7 marks*	3
Question 3	*5 marks*	5
Question 4	*8 marks*	6
Question 5	*7 marks*	8
Question 6	*10 marks*	10
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Question 9	*15 marks*	16
Question 10	*14 marks*	19
Question 11	*23 marks*	21

1. A puck of mass 0.40 kg moves on a smooth horizontal surface.
The horizontal unit vectors **i** and **j** are perpendicular.

A hockey stick strikes the puck.
Immediately before contact, the puck has velocity $9\mathbf{i}\,\mathrm{m\,s^{-1}}$.
Immediately after contact, its velocity is $12\mathbf{j}\,\mathrm{m\,s^{-1}}$.

(a) Find the impulse exerted by the stick on the puck, giving your answer as a vector, and calculate its magnitude. **[3]**

(b) During contact, the force exerted by the stick acts in a constant horizontal direction.
Its magnitude, F newtons, is modelled by

$$F = k\sin(25\pi t), \qquad 0 \le t \le 0.040$$

where k is a positive constant and t is the time in seconds measured from the start of contact.
The stick loses contact with the puck at $t = 0.040$.
Using your answer to part (a), find the exact value of k. **[3]**

Solution

(a) The impulse–momentum principle is

$$\mathbf{I} = m\mathbf{v} - m\mathbf{u}$$

Substituting the velocities gives the impulse exerted by the stick:

$$\begin{aligned}\mathbf{I} &= 0.40(12\mathbf{j}) - 0.40(9\mathbf{i}) \\ &= (-3.6\mathbf{i} + 4.8\mathbf{j})\,\mathrm{N\,s}\end{aligned}$$

Since **i** and **j** are perpendicular, its magnitude is

$$|\mathbf{I}| = \sqrt{(-3.6)^2 + 4.8^2} = 6\,\mathrm{N\,s}$$

(b) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

The force acts in a constant direction and F is its magnitude, so this integral over the contact interval is the magnitude of the impulse. Using the magnitude from part (a):

$$\begin{aligned}6 &= \int_0^{0.040} k\sin(25\pi t)\,\mathrm{d}t \\ &= \left[-\frac{k}{25\pi}\cos(25\pi t)\right]_0^{0.040} \\ &= \frac{k}{25\pi}\big(\cos 0 - \cos\pi\big) \\ &= \frac{2k}{25\pi}\end{aligned}$$

Rearranging gives

$$k = 75\pi$$

2. A trolley of mass 6 kg moves along a smooth horizontal track at $4\,\mathrm{m\,s^{-1}}$ and strikes a buffer. The buffer brings the trolley to rest without causing it to rebound. Air resistance is negligible.

(a) Find the magnitude of the impulse exerted by the buffer on the trolley. **[2]**

(b) The trolley first touches the buffer at time $t = 0$ and comes to rest at time $t = T$ seconds. During this interval, the buffer exerts a horizontal force opposite to the trolley's motion. The magnitude of this force is

$$F(t) = 18\,000t(T - t)\,\mathrm{N}, \qquad 0 \leq t \leq T$$

Using your answer to part (a), find the stopping time T and the maximum magnitude of the force exerted by the buffer. **[5]**

Solution

(a) Take the trolley's initial direction of motion as positive. The impulse–momentum principle is

$$I = mv - mu$$

Substituting the mass and velocities:

$$I = 6(0) - 6(4) = -24\,\mathrm{N\,s}$$

The magnitude of the impulse is 24 N s.

(b) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

The buffer force acts in a constant direction and $F(t)$ is its magnitude, so this integral from $t = 0$ to $t = T$ is the magnitude of the impulse. Using part (a):

$$\begin{aligned} 24 &= \int_0^T 18\,000t(T - t)\,\mathrm{d}t \\ &= 18\,000 \int_0^T (Tt - t^2)\,\mathrm{d}t \\ &= 18\,000\left[\frac{Tt^2}{2} - \frac{t^3}{3}\right]_0^T \\ &= 18\,000\left(\frac{T^3}{2} - \frac{T^3}{3}\right) \\ &= 3000T^3 \end{aligned}$$

Therefore the stopping time is

$$T = \sqrt[3]{\frac{24}{3000}} = 0.2\,\mathrm{s}$$

To find the maximum force, differentiate with respect to t:

$$\frac{\mathrm{d}F}{\mathrm{d}t} = 18\,000(T - 2t)$$

$$\frac{\mathrm{d}F}{\mathrm{d}t} = 0 \quad \Longrightarrow \quad t = \frac{T}{2} = 0.1\,\mathrm{s}$$

Since $\frac{\mathrm{d}^2F}{\mathrm{d}t^2} = -36\,000 < 0$, this gives the maximum force:

$$F_{\max} = 18\,000(0.1)(0.2 - 0.1) = 180\,\mathrm{N}$$

3. A sled of mass 20 kg slides down a smooth straight slope inclined at 30° to the horizontal.
It reaches a hydraulic buffer with speed $3\,\mathrm{m\,s^{-1}}$.

The buffer exerts a force on the sled directed up the slope.
When the buffer has been compressed by x metres, the magnitude of this force is F newtons, where

$$F = \frac{A}{1+2x} + 200x$$

and A is a positive constant.
The sled comes to rest after compressing the buffer by 0.5 m.
Throughout the compression, the sled moves along the slope.
Model the sled as a particle.

(a) Find, in terms of A, the work done against the buffer during this compression. **[2]**

(b) Using work and energy, find the exact value of A, taking account of the work done by gravity. **[3]**

Solution

(a) The work done by a variable force is

$$W = \int_{x_0}^{x_1} F\,\mathrm{d}x$$

The buffer force opposes the motion, so the work done against it is this integral of its magnitude F. Substituting the buffer force and compression limits:

$$\begin{aligned} W &= \int_0^{0.5} \left(\frac{A}{1+2x} + 200x\right) \mathrm{d}x \\ &= \left[\frac{A}{2}\ln(1+2x) + 100x^2\right]_0^{0.5} \\ &= \left(\frac{A}{2}\ln 2 + 25\right)\mathrm{J} \end{aligned}$$

(b) Take $g = 9.8\,\mathrm{m\,s^{-2}}$ and the final position as the zero level for gravitational potential energy. The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

The sled falls through a vertical height of $0.5\sin 30°$ metres. The buffer opposes the motion, so its work is the negative of the answer to part (a). The normal reaction does no work. Hence

$$\begin{aligned} \frac{1}{2}(20)(3)^2 + (20)(9.8)(0.5\sin 30°) - \left(\frac{A}{2}\ln 2 + 25\right) &= 0 \\ 90 + 49 - \frac{A}{2}\ln 2 - 25 &= 0 \\ \frac{A}{2}\ln 2 &= 114 \\ A &= \frac{228}{\ln 2} \end{aligned}$$

Final answers with $g = 10$: (b) $A = \frac{230}{\ln 2}$

4. A trolley of mass 80 kg travels along a straight horizontal track.
A powered launcher exerts a variable driving force on the trolley in the direction of motion for 2 s.
Throughout this interval, a constant resistance of 40 N acts against the motion.

At time $t = 0$, when the launcher begins to act, the trolley has speed $3\,\mathrm{m\,s^{-1}}$.
At time $t = 2\,\mathrm{s}$, its speed is $7\,\mathrm{m\,s^{-1}}$.
The trolley does not change its direction of motion.

(a) Find the magnitude of the impulse exerted on the trolley by the launcher during this interval. **[3]**

(b) The driving force exerted by the launcher is modelled by

$$F = kt^2(2 - t), \qquad 0 \leq t \leq 2$$

where F is measured in newtons, t is measured in seconds and k is a positive constant.

(i) Find the value of k. **[3]**

(ii) Determine the maximum driving force predicted by this model. **[2]**

Solution

(a) Take the direction of motion as positive. The impulse–momentum principle is

$$I_{\text{net}} = mv - mu$$

The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F \,\mathrm{d}t$$

Let I_L be the impulse exerted by the launcher. The resistance acts in the negative direction, so

$$I_L + \int_0^2 (-40)\,\mathrm{d}t = 80(7 - 3)$$

$$I_L - 80 = 320$$

$$I_L = 400\,\mathrm{N\,s}$$

(b) (i) The launcher's impulse is given by

$$I_L = \int_0^2 F \,\mathrm{d}t$$

Using part (a) and substituting the force model:

$$\begin{aligned} 400 &= \int_0^2 kt^2(2-t)\,\mathrm{d}t \\ &= k\int_0^2 (2t^2 - t^3)\,\mathrm{d}t \\ &= k\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^2 \\ &= k\left(\frac{16}{3} - 4\right) \\ &= \frac{4k}{3} \\ k &= 300 \end{aligned}$$

(ii) Differentiate the driving force:

$$\frac{\mathrm{d}F}{\mathrm{d}t} = 300(4t - 3t^2) = 300t(4 - 3t)$$

The derivative is positive for $0 < t < \frac{4}{3}$ and negative for $\frac{4}{3} < t < 2$. Since $F(0) = F(2) = 0$, the maximum occurs at $t = \frac{4}{3}$ s:

$$F_{\max} = 300\left(\frac{4}{3}\right)^2\left(2 - \frac{4}{3}\right) = \frac{3200}{9}\,\mathrm{N} \approx 356\,\mathrm{N}$$

5. A particle P of mass $2\,\text{kg}$ moves along a straight horizontal line.
Take the positive direction along the line to be the direction of its initial motion.

At time $t = 0$, the velocity of P is $3\,\text{m s}^{-1}$.
For $0 \leq t \leq 2$, the only horizontal force acting on P has signed component

$$F(t) = (3t^2 - 12t + 9)\,\text{N}$$

in the positive direction.
The particle continues to move in the positive direction throughout this interval.

(a) Find the impulse of the force over the interval $0 \leq t \leq 2$. **[2]**

(b) Find the speed of P at $t = 2$. **[2]**

(c) Find the work done by the force during the portion of the interval $0 \leq t \leq 2$ for which the force opposes the motion of P. **[3]**

Solution

(a) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

Substituting the force and time limits:

$$\begin{aligned} I &= \int_0^2 (3t^2 - 12t + 9)\,\mathrm{d}t \\ &= \left[t^3 - 6t^2 + 9t\right]_0^2 \\ &= 2\,\text{N s} \end{aligned}$$

(b) The impulse–momentum principle is

$$I = mv - mu$$

Using the impulse from part (a), and writing v_2 for the velocity at $t = 2$:

$$2 = 2v_2 - 2(3)$$

$$v_2 = 4\,\text{m s}^{-1}$$

Since the motion is in the positive direction, the speed is $4\,\text{m s}^{-1}$.

(c) Since the particle moves in the positive direction, the force opposes its motion when $F(t) < 0$. The zeros of the force satisfy

$$3t^2 - 12t + 9 = 0 \quad \Longrightarrow \quad t = 1 \text{ or } t = 3$$

The quadratic is negative between these roots, so the required portion is $1 < t \leq 2$. To find the velocity v_1 at $t = 1$, use the impulse formula:

$$\begin{aligned} I_{01} &= \int_0^1 F(t)\,\mathrm{d}t \\ &= \int_0^1 (3t^2 - 12t + 9)\,\mathrm{d}t \\ &= \left[t^3 - 6t^2 + 9t\right]_0^1 = 4\,\mathrm{N\,s} \end{aligned}$$

Using $I = mv - mu$:

$$\begin{aligned} 4 &= 2v_1 - 2(3) \\ v_1 &= 5\,\mathrm{m\,s^{-1}} \end{aligned}$$

The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

There is no change in gravitational potential energy. Let W be the work done by the horizontal force from $t = 1$ to $t = 2$; it is negative because the force opposes the motion. Using the speed from part (b):

$$\begin{aligned} \frac{1}{2}(2)(5^2) + W &= \frac{1}{2}(2)(4^2) \\ W &= 16 - 25 \\ &= -9\,\mathrm{J} \end{aligned}$$

6. A particle P of mass $4\,\text{kg}$ moves along a straight line on a smooth horizontal surface.
At time $t = 0$, its velocity is $4\,\text{m s}^{-1}$ in the positive direction.
The only horizontal force acting on P has component

$$F(t) = (3t^2 - 12t)\,\text{N}$$

in the positive direction, where t is the time in seconds and $t \geq 0$.

(a) Find the magnitude of the impulse exerted by the force between $t = 0$ and $t = 2$. **[3]**

(b) Find the speed of P at $t = 3$, and state whether it is moving in the positive or negative direction. **[3]**

(c) Find the exact time at which P next comes to rest after $t = 2$. **[4]**

Solution

(a) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

Substituting the force and time limits:

$$\begin{aligned} I &= \int_0^2 (3t^2 - 12t)\,\mathrm{d}t \\ &= \left[t^3 - 6t^2\right]_0^2 \\ &= -16\,\text{N s} \end{aligned}$$

The magnitude of the impulse is $16\,\text{N s}$.

(b) The impulse–momentum principle is

$$I = mv - mu$$

Over the interval from 0 to t, the impulse is $\int_0^t F(\tau)\,\mathrm{d}\tau$, writing τ for the variable of integration. Substituting $m = 4$ and $u = 4$:

$$\begin{aligned} 4v - 16 &= \int_0^t (3\tau^2 - 12\tau)\,\mathrm{d}\tau \\ &= \left[\tau^3 - 6\tau^2\right]_0^t \\ &= t^3 - 6t^2 \\ 4v &= 16 + t^3 - 6t^2 \end{aligned}$$

At $t = 3$:

$$v = \frac{16 + 3^3 - 6(3^2)}{4} = -\frac{11}{4}\,\text{m s}^{-1}$$

The speed is $\frac{11}{4}\,\text{m s}^{-1}$, and P is moving in the negative direction.

(c) Using the velocity relation from part (b), P is at rest when $v = 0$:

$$t^3 - 6t^2 + 16 = 0$$

$$(t-2)(t^2 - 4t - 8) = 0$$

Hence either $t = 2$, or

$$t^2 - 4t - 8 = 0$$

$$(t-2)^2 = 12$$

$$t = 2 \pm 2\sqrt{3}$$

The only root greater than 2 gives the next time at rest:

$$t = \left(2 + 2\sqrt{3}\right) \text{ s}$$

7. A particle of mass 4 kg moves along a straight horizontal line.
A fixed positive direction is chosen along the line.
At time t seconds, where $t \geq 0$, the component of the resultant force in this direction is

$$F(t) = 6(t-1)(t-3)\,\text{N}$$

At time $t = 0$, the velocity of the particle is $-1\,\text{m s}^{-1}$.

(a) Find the magnitude of the impulse exerted on the particle between $t = 0$ and $t = 1$. **[3]**

(b) Find the velocity of the particle at $t = 1$. **[2]**

(c) Hence find all times $t > 0$ at which the speed of the particle is $1\,\text{m s}^{-1}$. **[6]**

Solution

(a) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

Substituting the force and time limits:

$$\begin{aligned} I &= \int_0^1 6(t-1)(t-3)\,\mathrm{d}t \\ &= \int_0^1 (6t^2 - 24t + 18)\,\mathrm{d}t \\ &= \left[2t^3 - 12t^2 + 18t\right]_0^1 \\ &= 8\,\text{N s} \end{aligned}$$

The magnitude of the impulse is 8 N s.

(b) The impulse–momentum principle is

$$I = mv - mu$$

The impulse in part (a) is positive, so

$$\begin{aligned} 8 &= 4v - 4(-1) \\ v &= 1\,\text{m s}^{-1} \end{aligned}$$

(c) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$\begin{aligned} 4\frac{\mathrm{d}v}{\mathrm{d}t} &= 6(t-1)(t-3) \\ \frac{\mathrm{d}v}{\mathrm{d}t} &= \frac{3}{2}t^2 - 6t + \frac{9}{2} \end{aligned}$$

Integrating with respect to time:

$$v = \int \left(\frac{3}{2}t^2 - 6t + \frac{9}{2} \right) \mathrm{d}t$$

$$= \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t + c$$

Using $v = 1$ when $t = 1$, from part (b):

$$1 = \frac{1}{2} - 3 + \frac{9}{2} + c$$

$$c = -1$$

Thus the velocity is

$$v = \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t - 1$$

A speed of $1\,\mathrm{m\,s^{-1}}$ means that either $v = 1$ or $v = -1$.

For $v = 1$:

$$\frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t - 1 = 1$$

$$t^3 - 6t^2 + 9t - 4 = 0$$

$$(t-1)(t^2 - 5t + 4) = 0$$

$$(t-1)^2(t-4) = 0$$

$$t = 1 \text{ or } t = 4$$

For $v = -1$:

$$\frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t - 1 = -1$$

$$t^3 - 6t^2 + 9t = 0$$

$$t(t-3)^2 = 0$$

$$t = 0 \text{ or } t = 3$$

Excluding $t = 0$, the required times are

$$t = 1\,\mathrm{s}, \quad 3\,\mathrm{s}, \quad 4\,\mathrm{s}$$

8. In this question use $g = 9.8\,\mathrm{m\,s^{-2}}$.

A compact load of mass 60 kg is raised vertically by a cable.
When the load is x metres above its initial position, the tension in the cable is P newtons, where

$$P = 498 + \frac{k}{(x+2)^2}$$

The constant k satisfies $k > 360$.
The load starts from rest at $x = 0$, and the cable remains taut throughout the motion considered.
Air resistance is ignored.

(a) Find, in terms of k and h, the work done by the tension as the load rises from $x = 0$ to $x = h$. Also find, in terms of k, the height at which the tension equals the weight of the load. **[5]**

(b) The load has speed $3\,\mathrm{m\,s^{-1}}$ when $x = 6$. Determine k and find the speed of the load when $x = 8$, giving the speed to three significant figures. **[5]**

(c) Find the height at which the load first comes instantaneously to rest after leaving its initial position. **[2]**

(d) Give one physical reason why neglecting air resistance is reasonable for this load during its ascent. **[1]**

Solution

(a) The work done by a variable force is

$$W = \int_{x_0}^{x_1} F\,\mathrm{d}x$$

The tension acts in the direction of motion, so the work done by the tension is

$$\begin{aligned} W &= \int_0^h \left(498 + \frac{k}{(x+2)^2}\right)\mathrm{d}x \\ &= \left[498x - \frac{k}{x+2}\right]_0^h \\ &= \left(498h + k\left(\frac{1}{2} - \frac{1}{h+2}\right)\right)\mathrm{J} \end{aligned}$$

The weight is $60(9.8) = 588\,\mathrm{N}$. At the required height, $x \geq 0$, so $x + 2 > 0$. Equating the tension to the weight gives

$$\begin{aligned} 498 + \frac{k}{(x+2)^2} &= 588 \\ \frac{k}{(x+2)^2} &= 90 \\ (x+2)^2 &= \frac{k}{90} \\ x &= \left(\sqrt{\frac{k}{90}} - 2\right)\mathrm{m} \end{aligned}$$

(b) The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Take the initial position as the zero level for gravitational potential energy. During the ascent, the work done by the tension is positive; gravity is accounted for by the increase in gravitational potential energy. If the speed at height x is v, part (a) gives

$$0 + 0 + \left[498x + k\left(\frac{1}{2} - \frac{1}{x+2}\right)\right] = \frac{1}{2}(60)v^2 + 588x$$

$$30v^2 = -90x + k\left(\frac{1}{2} - \frac{1}{x+2}\right)$$

Substituting $x = 6$ and $v = 3$:

$$30(3)^2 = -90(6) + k\left(\frac{1}{2} - \frac{1}{8}\right)$$

$$270 = -540 + \frac{3k}{8}$$

$$\frac{3k}{8} = 810$$

$$k = 2160$$

At $x = 8$, taking the positive square root for the speed:

$$30v^2 = -90(8) + 2160\left(\frac{1}{2} - \frac{1}{10}\right)$$

$$= 144$$

$$v = \sqrt{\frac{24}{5}} \approx 2.19\,\text{m}\,\text{s}^{-1} \qquad (3\text{ s.f.})$$

(c) Using the energy relation in part (b), the speed is zero at height h when

$$0 = -90h + 2160\left(\frac{1}{2} - \frac{1}{h+2}\right)$$

$$= h\left(-90 + \frac{1080}{h+2}\right)$$

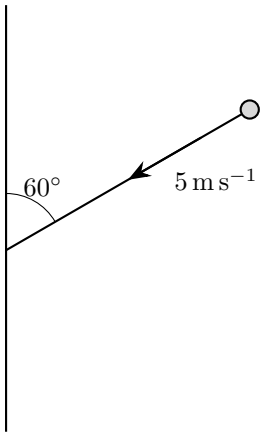
The initial position $h = 0$ is excluded, so

$$\frac{1080}{h+2} = 90$$

$$h + 2 = 12$$

$$h = 10\,\text{m}$$

(d) Its compact shape gives a small frontal area relative to its mass, so air resistance is small compared with its weight.



9. A ball of mass 0.4 kg moves on a smooth horizontal surface and collides with a smooth fixed vertical wall. Immediately before impact, its speed is $5\,\mathrm{m\,s^{-1}}$ and its direction makes an angle of $60°$ with the wall, as shown in the diagram above.

The ball first touches the wall at time $t = 0$ and loses contact at time $t = 0.2\,\mathrm{s}$.
During contact, the magnitude of the force exerted by the wall on the ball is modelled by

$$F = \begin{cases} kt(0.2 - t), & 0 \leq t \leq 0.1 \\ k(0.2 - t)^2, & 0.1 < t \leq 0.2 \end{cases}$$

where F is measured in newtons and k is a positive constant.

(a) Show, by integration, that the magnitude of the impulse exerted by the wall during the collision is

$$\frac{k}{1000}\,\mathrm{N\,s} \qquad \textbf{[5]}$$

(b) The coefficient of restitution between the ball and the wall is known to lie between $\frac{1}{4}$ and $\frac{3}{4}$, inclusive. Determine the corresponding range of values of k, giving exact bounds and fully justifying your answer. **[5]**

(c) Given that $k = 1500\sqrt{3}$, find the speed of the ball immediately after the collision in exact form, and the angle its direction of motion makes with the wall, to the nearest $0.1°$. **[5]**

Solution

(a) The impulse of a variable force is

$$I = \int_{t_0}^{t_1} F\,\mathrm{d}t$$

The wall's force acts in a constant direction and F is its magnitude, so this integral from $t = 0$ to $t = 0.2$ is the magnitude of the impulse. Splitting the integral at $t = 0.1$ and expanding the integrands:

$$\begin{aligned} I &= k\int_0^{0.1} (0.2t - t^2)\,\mathrm{d}t + k\int_{0.1}^{0.2} (0.04 - 0.4t + t^2)\,\mathrm{d}t \\ &= k\left[0.1t^2 - \frac{t^3}{3}\right]_0^{0.1} + k\left[0.04t - 0.2t^2 + \frac{t^3}{3}\right]_{0.1}^{0.2} \\ &= \frac{k}{1500} + \frac{k}{3000} \\ &= \frac{k}{1000}\,\mathrm{N\,s} \end{aligned}$$

(b) Take the direction away from the wall as positive. Let u and v be the signed velocity components perpendicular to the wall before and after impact, respectively. Then

$$u = -5\sin 60^\circ = -\frac{5\sqrt{3}}{2}\,\mathrm{m\,s^{-1}}$$

Writing the coefficient of restitution as e, Newton's law of restitution gives

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} = \frac{v}{-u}$$

$$v = -eu = \frac{5\sqrt{3}}{2}e$$

The impulse–momentum principle, applied perpendicular to the wall, is

$$I = mv - mu$$

Using part (a):

$$\begin{aligned} \frac{k}{1000} &= 0.4\left(\frac{5\sqrt{3}}{2}e\right) - 0.4\left(-\frac{5\sqrt{3}}{2}\right) \\ &= \sqrt{3}(e+1) \\ k &= 1000\sqrt{3}(e+1) \end{aligned}$$

Since $1000\sqrt{3} > 0$, k increases as e increases. Thus the inclusive bounds $\frac{1}{4} \le e \le \frac{3}{4}$ give

$$1000\sqrt{3}\left(1 + \frac{1}{4}\right) \le k \le 1000\sqrt{3}\left(1 + \frac{3}{4}\right)$$

$$1250\sqrt{3} \le k \le 1750\sqrt{3}$$

(c) Substituting $k = 1500\sqrt{3}$ into the relation from part (b):

$$\begin{aligned} 1500\sqrt{3} &= 1000\sqrt{3}(e+1) \\ e + 1 &= \frac{3}{2} \\ e &= \frac{1}{2} \end{aligned}$$

The wall is smooth, so there is no impulse parallel to it and the parallel velocity component is unchanged. The rebound components are therefore

$$v = \frac{5\sqrt{3}}{2} \times \frac{1}{2} = \frac{5\sqrt{3}}{4}\,\mathrm{m\,s^{-1}}$$

$$v_{\parallel} = 5\cos 60^\circ = \frac{5}{2}\,\mathrm{m\,s^{-1}}$$

Let V be the rebound speed and θ the angle its direction makes with the wall. Using Pythagoras and the ratio of the perpendicular and parallel components:

$$V^2 = \left(\frac{5\sqrt{3}}{4}\right)^2 + \left(\frac{5}{2}\right)^2 = \frac{75}{16} + \frac{100}{16} = \frac{175}{16}$$

$$\tan\theta = \frac{v}{v_{\parallel}} = \frac{5\sqrt{3}/4}{5/2} = \frac{\sqrt{3}}{2}$$

Hence the speed and angle, with the angle rounded to the nearest 0.1°, are

$$V = \frac{5\sqrt{7}}{4}\,\mathrm{m\,s^{-1}}, \qquad \theta = \arctan\left(\frac{\sqrt{3}}{2}\right) \approx 40.9^\circ$$

10. A car of mass 800 kg moves along a straight horizontal road.
At time $t = 0$, it passes a marker with speed $20\,\mathrm{m\,s^{-1}}$.
After travelling a distance x m beyond the marker, its speed is $v\,\mathrm{m\,s^{-1}}$.

The driving force varies with distance and has magnitude

$$\left(1800 + 1200e^{-x/200}\right)\ \mathrm{N}$$

The resistance to motion has magnitude $2v^2$ N.

(a) Form a differential equation relating v and x.
By making the substitution $u = v^2$, solve this equation to show that

$$v^2 = 900 + (3x - 500)e^{-x/200}$$ **[8]**

(b) Using the work–energy principle and your result from part (a), find the exact work done against resistance during the first 200 m beyond the marker.
Hence find the average magnitude of the resistance over this distance. **[6]**

Solution

(a) Take the direction of motion as positive. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$800v\frac{\mathrm{d}v}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2v^2$$

With the substitution $u = v^2$:

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2v\frac{\mathrm{d}v}{\mathrm{d}x}$$

Hence the differential equation becomes

$$400\frac{\mathrm{d}u}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2u$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{u}{200} = \frac{9}{2} + 3e^{-x/200}$$

The integrating factor is

$$e^{\int \frac{1}{200}\,\mathrm{d}x} = e^{x/200}$$

Multiplying the equation by this factor gives

$$e^{x/200}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{1}{200}e^{x/200}u = \frac{9}{2}e^{x/200} + 3$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(ue^{x/200}\right) = \frac{9}{2}e^{x/200} + 3$$

Integrating with respect to x:

$$ue^{x/200} = 900e^{x/200} + 3x + c$$

$$u = 900 + (3x + c)e^{-x/200}$$

At the marker, $x = 0$ and $u = 20^2 = 400$, so

$$400 = 900 + c$$

$$c = -500$$

Replacing u by v^2:

$$v^2 = 900 + (3x - 500)e^{-x/200}$$

(b) The work done by a variable force is

$$W = \int_{x_0}^{x_1} F \, \mathrm{d}x$$

The driving force acts in the direction of motion, so the work done by the driving force over the first 200 m is

$$\begin{aligned} W_D &= \int_0^{200} \left(1800 + 1200e^{-x/200}\right) \mathrm{d}x \\ &= \left[1800x - 240\,000e^{-x/200}\right]_0^{200} \\ &= \left(600\,000 - \frac{240\,000}{e}\right) \text{J} \end{aligned}$$

From part (a), the speed at $x = 200$ satisfies

$$v^2 = 900 + (600 - 500)e^{-1} = 900 + \frac{100}{e}$$

The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Let W_R be the positive work done against resistance. The work done by resistance is $-W_R$, and there is no change in gravitational potential energy on the horizontal road. Thus

$$\begin{aligned} \frac{1}{2}(800)(20^2) + W_D - W_R &= \frac{1}{2}(800)\left(900 + \frac{100}{e}\right) \\ W_R &= W_D - 400\left(500 + \frac{100}{e}\right) \\ &= 600\,000 - \frac{240\,000}{e} - 200\,000 - \frac{40\,000}{e} \\ &= \left(400\,000 - \frac{280\,000}{e}\right) \text{J} \end{aligned}$$

The average force over a displacement is

$$F_{\text{av}} = \frac{W}{x_1 - x_0}$$

Hence the average resistance over the first 200 m is

$$\begin{aligned} R_{\text{av}} &= \frac{400\,000 - \frac{280\,000}{e}}{200} \\ &= \left(2000 - \frac{1400}{e}\right) \text{N} \end{aligned}$$

11. A particle of mass 2 kg moves along a horizontal straight line.
At time $t = 0$, it passes through a point O with velocity $1\,\mathrm{m\,s^{-1}}$.
The resultant force acts in the direction of motion and has magnitude F N, where

$$F = \frac{27}{v} - 3v$$

and $v\,\mathrm{m\,s^{-1}}$ is the velocity of the particle.
Its displacement from O at time t seconds is x m.

(a) Formulate and solve a differential equation relating v and x to show that

$$x = \ln\left(\frac{3+v}{2(3-v)}\right) - \frac{2}{3}(v-1), \qquad 1 \leq v < 3$$

[7]

(b) Find the exact displacement of the particle when its velocity first reaches $2\,\mathrm{m\,s^{-1}}$.
By changing the variable of integration from x to v, calculate the work done by the force up to this instant.
Hence find the average force over this displacement, giving your answer to three significant figures. **[7]**

(c) Formulate and solve a differential equation relating v and t to obtain an expression for v in terms of t. **[5]**

(d) Calculate the velocities at $t = 1$ and $t = 2$, giving your answers to four significant figures.
Find the impulse of the force over the interval $1 \leq t \leq 2$, giving your answer to three significant figures. **[4]**

Solution

(a) Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{27}{v} - 3v$$

$$\frac{\mathrm{d}x}{\mathrm{d}v} = \frac{2v^2}{3(9-v^2)}$$

To integrate, rearrange and use partial fractions:

$$\frac{2v^2}{3(9-v^2)} = -\frac{2}{3} + \frac{6}{9-v^2}$$

$$= -\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}$$

Separating variables and integrating:

$$\int \mathrm{d}x = \int \left(-\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}\right)\mathrm{d}v$$

$$x = -\frac{2}{3}v - \ln|3-v| + \ln|3+v| + c$$

At O, $x = 0$ and $v = 1$, so

$$0 = -\frac{2}{3} - \ln 2 + \ln 4 + c$$

$$c = \frac{2}{3} - \ln 2$$

The force is positive for $1 \leq v < 3$, so the velocity increases towards 3. The logarithmic term tends to infinity as $v \to 3^-$, so $v = 3$ is not reached at any finite displacement. Substituting the constant and combining the logarithms gives

$$x = \ln\left(\frac{3+v}{2(3-v)}\right) - \frac{2}{3}(v-1), \qquad 1 \leq v < 3$$

(b) Let s be the displacement when $v = 2$. From part (a):

$$s = \left(\ln\left(\frac{5}{2}\right) - \frac{2}{3}\right) \text{m}$$

The work done by a variable force is

$$W = \int_{x_0}^{x_1} F \, \mathrm{d}x$$

The force acts in the direction of motion, so the work it does is this integral of its magnitude F. Using $\frac{\mathrm{d}x}{\mathrm{d}v}$ from part (a) to change the variable of integration:

$$\begin{aligned} W &= \int_0^s \left(\frac{27}{v} - 3v\right) \mathrm{d}x \\ &= \int_1^2 \left(\frac{27}{v} - 3v\right) \frac{2v^2}{3(9-v^2)} \, \mathrm{d}v \\ &= \int_1^2 \frac{3(9-v^2)}{v} \frac{2v^2}{3(9-v^2)} \, \mathrm{d}v \\ &= \int_1^2 2v \, \mathrm{d}v \\ &= \left[v^2\right]_1^2 = 3 \, \text{J} \end{aligned}$$

The average force over a displacement is

$$F_{\text{av}} = \frac{W}{x_1 - x_0}$$

Hence

$$F_{\text{av}} = \frac{3}{\ln\left(\frac{5}{2}\right) - \frac{2}{3}} \approx 12.0 \, \text{N} \qquad (3 \text{ s.f.})$$

(c) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{27}{v} - 3v$$

$$2v\frac{\mathrm{d}v}{\mathrm{d}t} = 27 - 3v^2$$

Separating variables and integrating, with $1 \leq v < 3$:

$$\int \frac{2v}{9 - v^2}\,\mathrm{d}v = \int 3\,\mathrm{d}t$$

$$-\ln(9 - v^2) = 3t + k$$

Using $v = 1$ when $t = 0$ gives $k = -\ln 8$. Therefore

$$-\ln(9 - v^2) = 3t - \ln 8$$

$$\ln\left(\frac{9 - v^2}{8}\right) = -3t$$

$$9 - v^2 = 8e^{-3t}$$

$$v^2 = 9 - 8e^{-3t}$$

Taking the positive square root, since the particle moves in the positive direction:

$$v = \sqrt{9 - 8e^{-3t}}, \qquad t \geq 0$$

(d) Substituting into the expression from part (c):

$$v(1) = \sqrt{9 - 8e^{-3}} \approx 2.933\,\mathrm{m\,s^{-1}} \qquad (4\text{ s.f.})$$

$$v(2) = \sqrt{9 - 8e^{-6}} \approx 2.997\,\mathrm{m\,s^{-1}} \qquad (4\text{ s.f.})$$

The impulse–momentum principle is

$$I = mv - mu$$

For this interval, the initial and final velocities are $v(1)$ and $v(2)$. Using their unrounded values:

$$I = 2\big[v(2) - v(1)\big]$$

$$= 2\left(\sqrt{9 - 8e^{-6}} - \sqrt{9 - 8e^{-3}}\right)$$

$$\approx 0.128\,\mathrm{N\,s} \qquad (3\text{ s.f.})$$